Week 9: Bayesian statistical inference

F79MA Statistical Models A

Heriot-Watt University

Bayesian Inference

Decision theory

In this unit we give a brief outline of Decision Theory and the related concepts such as loss (and utility). Decision theory has an important role in statistics which is sometimes known as 'decision-making under uncertainty'.

Elements of a decision problem

The simplest decision problems are finite and discrete and have the following components:

1. A set of possible *actions* (known as the action space):

$$A = \{a_1, a_2, \ldots, a_n\}.$$

   The decision is to select one of these actions.

2. A set of possible *states of nature* of which one will occur.

$$S = \{\theta_1, \theta_2, \ldots, \theta_m\}.$$

3. A *set of consequences* which is formed from the set of pairs $\{(\theta_i, a_j)\}$. Associated with each consequence we have a loss $L(\theta_i, a_j)$. This represents the loss that will be incurred if I choose action a_j when state θ_i is the case.

4. This leads to the idea of the *loss function* $L(\theta, a)$ which assigns to each consequence a loss that is incurred in the event that the consequence occurs. In the case of the finite, discrete problem this can be represented as a table or matrix $[L(\theta_i, a_j)]_{m\times k}$.

A very simple example: I roll a die and don't tell you the score (state of nature). You must guess the score (action). If you're right I pay you £5 – otherwise you pay me £1. In this case, $m = k$ and the loss matrix $[L(\theta_i, a_j)]_{m\times k}$ has elements $[L(\theta_i, a_j)]_{i,j} = -£5$ on the diagonal $i = j$, and $[L(\theta_i, a_j)]_{i,j} = £1$ otherwise.

Decision theory is all about how to choose the action in order to minimise the loss in some way. There are various strategies for making this choice. For example, one could follow a *minimax* approach and choose the action which minimises the maximum possible loss. This is a very risk-averse strategy that does not take into account the likelihood of this worse-case scenario.

In statistical decision theory, we seek to make use of further information on how likely the different scenarios or "states of nature" are. Suppose that we model θ as a random quantity and describe our uncertainty about θ using a probability function:

$$\pi(\theta_i), i = 1, 2, 3, \ldots, m.$$

This could come from prior knowledge or from a Bayesian analysis of some data x, in which case $\pi(\theta_i) = \pi(\theta_i|x)$. How one proceeds does not depend on which of these two approaches have led to $\pi(\theta)$. Then the Bayes principle is to select the action which minimises the expected loss, where the expected loss associated with action a_j is defined formally as:

$$B(a_j) = E_\theta(L(\theta, a_j)) = \sum_i L(\theta_i, a_j)\pi(\theta_i).$$

Bayesian Hypothesis Testing

For illustration, we now show how hypothesis testing can be formulated within the Bayesian decision theoretic framework.

Recall that, given data $\underline{x}$, the posterior distribution $\pi(\theta|\underline{x})$ contains all the information about θ provided by the data. Therefore, when it comes to testing hypotheses we can consider

$$\mathbb{P}(H_0 \text{ true}|\underline{x}) = \mathbb{P}(\theta \in \omega|\underline{x}) = \int_\omega \pi(\theta|\underline{x})d\theta.$$

Hypothesis testing is a form of decision problem where the actions are

- a_1 – do not reject H_0,
- a_2 – reject H_0.

and the states of nature are

- H_0 true,
- H_0 false.

To illustrate this approach, we consider two simple hypotheses (as we did when considering the NP-Lemma)

$$H_0 : \theta = \theta_0 \; v. \; H_1 : \theta = \theta_1.$$

We require a loss table

$$\begin{array}{ccc} & a_1 & a_2 \\ \theta_0 & L_{00} & L_{01} \\ \theta_1 & L_{10} & L_{11} \end{array}$$

We have prior probabilities $\pi(\theta_0)$ and $\pi(\theta_1)$. Then, given data $\underline{x}$, we obtain posterior probabilities $\pi(\theta_0|\underline{x})$ and $\pi(\theta_1|\underline{x})$ for our respective competing hypotheses.

The Bayes principle says we should select the action (to reject or not reject H_0) to minimise the expected posterior loss. This means we should select the smaller of:

$$\begin{aligned} B(a_1) &= L_{00}\pi(\theta_0|\underline{x}) + L_{10}\pi(\theta_1|\underline{x}) \\ B(a_2) &= L_{01}\pi(\theta_0|\underline{x}) + L_{11}\pi(\theta_1|\underline{x}) \end{aligned}$$

We often set $L_{00} = L_{11} = 0$ so that there is no loss if you select the correct option. Then

$$\begin{aligned} B(a_1) &= L_{10}\pi(\theta_1|\underline{x}) \\ B(a_2) &= L_{01}\pi(\theta_0|\underline{x}) \end{aligned}$$

Therefore we should select a_1 if

$$L_{10}\pi(\theta_1|\underline{x}) < L_{01}\pi(\theta_0|\underline{x})$$

i.e. if

$$\frac{\pi(\theta_0|\underline{x})}{\pi(\theta_1|\underline{x})} > \frac{L_{10}}{L_{01}}.$$

Since $\pi(\theta|\underline{x}) \propto \pi(\theta)f(\underline{x};\theta)$, this is equivalent to

$$\frac{\pi(\theta_0)f(\underline{x};\theta_0)}{\pi(\theta_1)f(\underline{x};\theta_1)} > \frac{L_{10}}{L_{01}}$$

or the likelihood ratio test

$$\frac{f(\underline{x};\theta_0)}{f(\underline{x};\theta_1)} > \frac{\pi(\theta_1)L_{10}}{\pi(\theta_0)L_{01}}.$$

As in the classical case we reject H_0 only if the likelihood ratio is too small! The key difference is the way in which we choose the threshold k. In the classical case we choose the value of k to fix the value of α under H_0. By contrast, in the Bayesian case we choose k depending on our priors beliefs and our losses.

Bayesian point estimation

We conclude this unit by showing how point estimation can be formulated within the Bayesian decision theoretic framework. This leads to the notion of Bayesian point estimators.

Let the state of nature θ be an unknown parameter or quantity of interest. Suppose that we have observed some data $\underline{x}$ and derived a posterior distribution $\pi(\theta|\underline{x})$ that also takes into account any prior knowledge about θ.

In point estimation, we need to choose what value $\hat{\theta}$ to report as estimate of $\theta|\underline{x}$, so the choice of $\hat{\theta}$ is in fact the action in our decision problem. The loss $L(\theta, a)$ represents the error associated with reporting a when the truth is θ. Given that the value of θ is unknown and that we are operating with the random quantity $\theta|\underline{x}$, we will naturally look at the error associated with summarising $\theta|\underline{x}$ with a single value (the reported estimate).

Indeed, *the Bayes Principle* says that we should choose the action, a (i.e. the value of our estimate $\hat{\theta}$) so as to minimise the expected loss. In our context, this reads:

$$B(a) = E_{\theta|\underline{x}}\{L(\theta, a)\} = \int L(\theta, a)\pi(\theta|\underline{x})\mathrm{d}\theta.$$

Accordingly, the Bayesian estimator of $\theta|\underline{x}$ w.r.t. L is the value of a that minimises the expected estimation error, as defined by the loss L. That is, $\hat{\theta} = \text{argmin}B(a)$.

Bayesian estimation w.r.t. the mean square error loss For illustration, suppose that $L(\theta, a) = (a-\theta)^2$ related to the squared estimation error. The expected loss is therefore

$$B(a) = \int (a-\theta)^2 \pi(\theta \mid \underline{x}) d\theta\,.$$

To minimise the expected loss we calculate

$$\frac{d}{da}B(a) = 2\int (a-\theta)\pi(\theta \mid \underline{x})d\theta.$$

Equating this to zero for minimum $\hat{\theta} = \text{argmin}\, B(a)$ leads to

$$\hat{\theta}\int \pi(\theta \mid \underline{x})d\theta = \hat{\theta} = \int \theta\pi(\theta \mid \underline{x})d\theta.$$

So, $\hat{\theta}$ is the *m*ean of the posterior distribution; that is, $\hat{\theta} = \mathbb{E}(\theta|\underline{x})$.

Example - Normal distribution Let $\underline{X}$ be a random sample from $\mathrm{N}(\mu, \sigma^2)$ distribution, with known σ^2. Let the prior distribution for μ be $\mathrm{N}(\mu_0, \sigma_0^2)$, with known μ_0 and σ_0^2. Calculate the Bayesian estimator of μ w.r.t. the mean square error loss.

Likelihood $\propto \exp\left(-\frac{n}{2\sigma^2}(\mu - \overline{x})^2\right)$.

Prior $\propto \exp\left(-\frac{1}{2\sigma_0^2}(\mu - \mu_0)^2\right)$.

We know that the posterior density is proportional to the product of prior density and likelihood. After some simple manipulations, one can show that the posterior is proportional to

$$\exp\left(-\frac{1}{2\sigma_1^2}(\mu - \mu_1)^2\right)$$

where

$$\mu_1 = \frac{\sigma_0^2\overline{x} + \frac{\sigma^2}{n}\mu_0}{\sigma_0^2 + \frac{\sigma^2}{n}} \quad \text{and} \quad \sigma_1^2 = \frac{\frac{\sigma^2}{n}\cdot\sigma_0^2}{\frac{\sigma^2}{n} + \sigma_0^2}.$$

So, the posterior distribution for the parameter μ is $\mathrm{N}(\mu_1, \sigma_1^2)$. We write this as

$$\mu|_{\underline{x}} \sim \mathrm{N}(\mu_1, \sigma_1^2).$$

Then, the Bayesian estimator w.r.t. to the square error loss is:

$$\widehat{\mu} = \frac{\sigma_0^2 \overline{X} + \frac{\sigma^2}{n}\mu_0}{\sigma_0^2 + \frac{\sigma^2}{n}},$$

which we can rewrite as:

$$\widehat{\mu} = \frac{\frac{1}{\sigma^2/n}}{\frac{1}{\sigma^2/n} + \frac{1}{\sigma_0^2}} \overline{X} + \frac{\frac{1}{\sigma_0^2}}{\frac{1}{\sigma^2/n} + \frac{1}{\sigma_0^2}} \mu_0.$$

Notice that this estimator is a *weighted average* of $\overline{X}$ (this is the classical MLE which depends on data only) and of μ_0 (this comes from the prior), with weights reflecting the relative confidence in observed and prior information.